Business cycles facts and methods

Advanced Macroeconomics

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March-April 2026

Organization

- This is my first time teaching this class
  - Feedback welcome
  - Tentative plan and readings on syllabus. Will likely update as we go.
- The class material is on github (`https://github.com/sara-casella/Luiss-AdvancedMacro-2026`)
  - Slides will be posted online ahead of class
  - There will be typos. Please point them out.
- Ask questions

Grade and Final Exam

The grade will consist of two components:

- 20% problem sets (limited review)
  - A mix of theory, empirics, numerical methods
- 80% exam

Let’s Start!

Outline

1. What are business cycles? Separating trend from cycle
    - Focus on filtering the data
2. Some stylized “facts”
3. Towards causal inference in Macroeconomics

The Trend and the Cycle

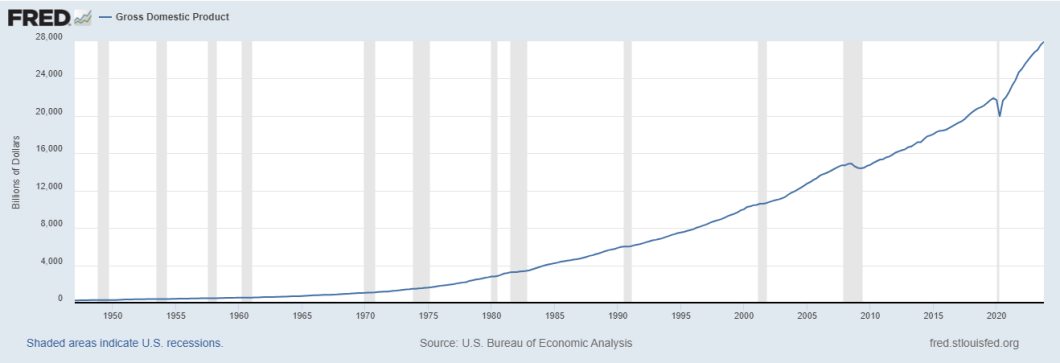
Two Approaches

- Economy fluctuates over time
- When we study cycles we want to separate them from longer-run movements
- Two approaches:
  - NBER approach: judgemental
  - Statistical: using a particular standardized method

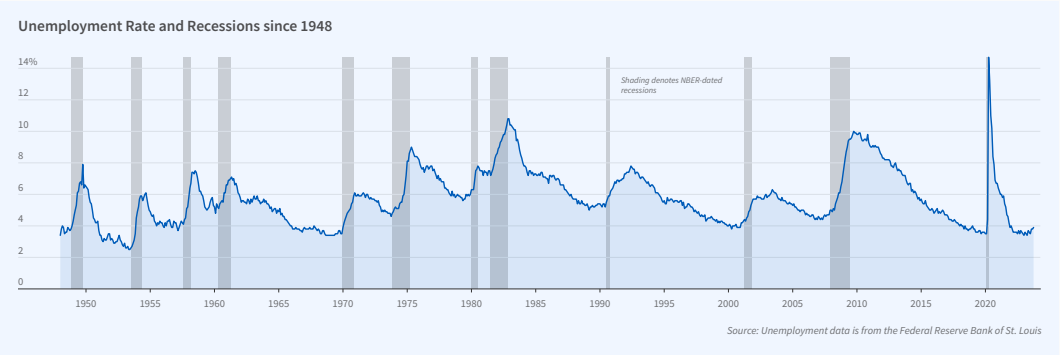
The NBER Approach

- NBER Business Cycle Dating Committee: Robert Hall, Robert Gordon, James Poterba, Valery Ramey, Christina Romer, David Romer, James Stock, Mark Watson
- "A recession is a significant decline in economic activity spread across the economy, lasting more than a few months, normally visible in real GDP, real income, employment, industrial production, and wholesale-retail sales. A recession begins just after the economy reaches a peak of activity and ends as the economy reaches its trough. Between trough and peak, the economy is in an expansion. Expansion is the normal state of the economy; most recessions are brief and they have been rare in recent decades"
- Europe VS US: `https://eabcn.org/sites/default/files/theceprandnberapproaches_0.pdf`

U.S. GDP (today USD 31 trillion)



U.S. Unemployment Rate



The Statistical Approach

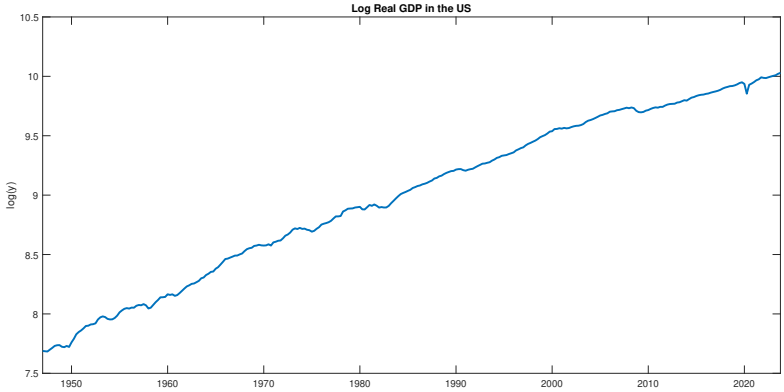
- Idea: non-stationary (trending) variables can be separated into a cyclical stationary part and a trending non-stationary
- Stationary part should have well-defined moments (mean, autocorrelation, standard deviation, etc.)
- Empirical problem: no unique way of separating cycle from trend
- A time series can be represented in different ways: a sum of sine waves of different frequencies (frequency domain) or a sequence of random variables correlated in certain ways (time domain). Each offers a way to remove trend.

Simple Example

Estimate a trend with constant growth rate (log-linear):

$$y_t = \hat{y}_0 + \hat{g}_y t + \hat{z}_t \quad (y \sim \log GDP)$$

Define the deviation z_t as the business cycle



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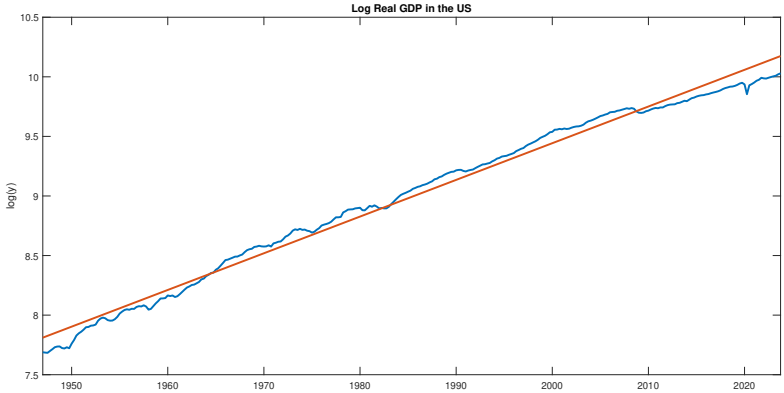

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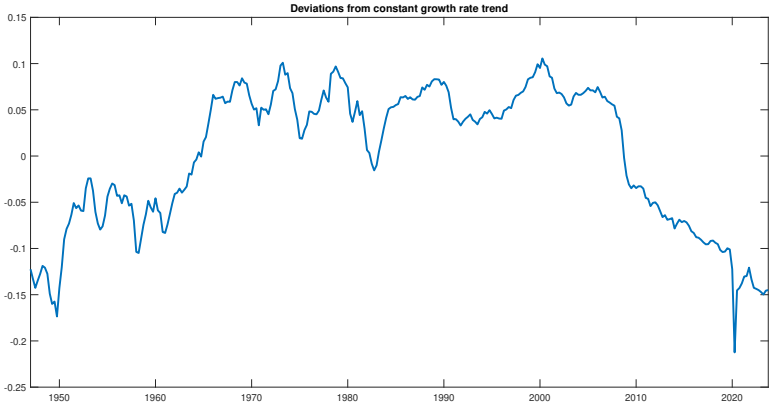


Hodrick-Prescott Filter

- Most common filter: solution to

$$\min_{\{\tau\}_{t-1}^{T}} \left\{ \sum_{t=1}^{T} (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2 \right\}$$

- Changes in the first-differences ("roughness") of the trend are penalized
- Higher λ gives smoother trend

Matrix Notation

- The first order condition is linear $\Rightarrow$ easy to implement
- For a given λ, the trend is obtained as

$$\tau = T^{-1} y$$

with

$$T \equiv \begin{bmatrix} \lambda+1 & -2\lambda & \lambda & 0 & 0 & & 0 & 0 & 0 & 0 & 0 \\ -2\lambda & 5\lambda+1 & -4\lambda & \lambda & 0 & \cdots & 0 & 0 & 0 & 0 & 0 \\ \lambda & -4\lambda & 6\lambda+1 & 4\lambda & \lambda & & 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda & -4\lambda & 6\lambda+1 & 4\lambda & & 0 & 0 & 0 & 0 & 0 \\ & & \vdots & & & & & & \vdots & & \\ 0 & 0 & 0 & 0 & 0 & & \lambda & -4\lambda & 6\lambda+1 & 4\lambda & \lambda \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & \lambda & -4\lambda & 5\lambda+1 & -2\lambda \\ 0 & 0 & 0 & 0 & 0 & & 0 & 0 & \lambda & -2\lambda & \lambda+1 \end{bmatrix}$$

- The cycle is $y - \tau$, linear combination of all previous and future values of y_t

HP Implementation

- It has become standard to use $\lambda = 1600$ for quarterly data.
- For yearly data, some use linear adjustment, implying ($\lambda = 1600/4 = 400$) some quadratic ($\lambda = 1600/16 = 100$) and some even fourth-order ($\lambda = 1600/4^4 = 6.25$) adjustment
- For a discussion see e.g., Marcet and Ravn, 2003; `https://econ-papers.upf.edu/papers/588.pdf`
- Some potential problems with HP; can retain too much low-frequency and/or high-frequency variation. Some fixes in the Hamilton filter (`https://www.nber.org/papers/w23429`). Still used a lot, however.
- Alternative: band-pass filter. I.e., decide on a spectrum, e.g., 3-8 years: frequency-domain method

Some Stylized Facts

Regularities

Given data that is filtered, we are interested in:

1. Volatility: volatility is measured by standard deviation of the detrended logged variable. Investment is much more volatile than output, consumption less volatile (smoothing).
2. Correlations: the correlation between the detrended logged variable and (tipically) detrended logged output
3. Persistence: the correlation between the detrended logged variable and its own lagged value

Definitions

- A variable x is pro-cyclical if $corr(y_{c,t}, x_{c,t} > 0)$ (investment, consumption)
- x is counter-cyclical if $corr(y_{c,t}, x_{c,t} < 0)$ (unemployment)
- x is leading if $corr(y_{c,t+s}, x_{c,t})$ is highest and positive for $s > 0$ (new housing starts)
- x is lagging if $corr(y_{c,t+s}, x_{c,t})$ is highest and positive for $s < 0$ (changes in CPI)

Some Typical Findings

Business Cycle Statistics - Annual U.S. Data

Variable-logged	S.D.	Correlation	Autocorrelation
Output	3.5	1.00	0.66
Consumption	2.2	0.74	0.72
Investment	10.5	0.68	0.25
Hours Worked	2.1	0.81	0.39
Productivity	2.2	0.82	0.77

Logged variable detrended using a linear-quadratic time trend.

Source: Greenwood et al. (1988)

Other facts:

- Durable consumption purchases very volatile and procyclical
- Most sectors positively correlated (comovement)
- Real wages only weakly procyclical (if at all)
- Profits highly procyclical, interests less so

Towards Causal Inference

Causal Inference in Macroeconomics

- Causal inference huge challenge
- Endogeneity is pervasive (e.g. Central Banks lower rates in bad times)
- External validity big issue
  - Short vs Long Run
  - Effect of policy state dependent (business cycle, response of other policies, savings, how open economy...)
  - Degree to which a policy action is a surprise can affect both how strongly and when the economy reacts to it
- As a results much work is structural in nature

Different empirical approaches for business cycle analysis

Different methods pursued in modern macro, historical order:

- Estimation of theoretical models, recovering "deep parameters" and thus avoiding Lucas critique. Identification from imposing specific model structure.
- Vector autoregressions (VARs): estimation of dynamic linear relationships
- Structural VARs (SVARs): identification like for VARs plus more theory (short and long run restrictions, sign restrictions)
- Local Projections: similar to VARs o SVARs
- Natural experiments? e.g. "Narrative approach" (Romer Romer, 1989)
- High-frequency identification (monetary policy).
- Cross-sectional identification (e.g., Bartik, granular instruments), labor, trade...

Local Projections: The "Direct" Approach to Impulse Responses

- **Definition:** Local Projections (LP) estimate impulse responses (IRFs) at each horizon h by regressing the outcome at $t + h$ directly on the intervention at t.
- **Theoretical Appeal:**
  - LPs are **semi-parametric**; they impose "mild" assumptions on the data and the shape of the response.
  - Nexus between macroeconomics and the micro-applied **policy evaluation** literature.
- **Key Advantages over VARs:**
  - **Robustness:** More robust to misspecification; small errors do not compound over horizons like they do in iterated VARs.
  - **Flexibility:** Simple to handle non-linearities and state-dependence.

Defining the Impulse Response

- The **Impulse Response** $R(h, \delta)$ is defined as the difference between two conditional expectations:

$$R(h, \delta) \equiv E[y_{t+h}|s_t = s_0 + \delta; x_t] - E[y_{t+h}|s_t = s_0; x_t]$$

- **Components:**
  - y_{t+h}: Outcome variable at horizon h.
  - s_t: The intervention, shock, or treatment.
  - x_t: Vector of exogenous and predetermined control variables.
  - δ: The "dose" or size of the intervention.
- **Linearity Assumptions:** In linear models, the response is proportional to the dose δ, and the baseline s_0 cancels out.

Baseline LP Estimation: LP-OLS

- For each horizon $h = 0, 1, \ldots, H$, estimate the regression:

$$y_{t+h} = \alpha_h + \beta_h s_t + \gamma_h x_t + v_{t+h}$$

- **Consistency:** $E[s_t v_{t+h}] = 0$.
- **Residual Structure:**
  - The error v_{t+h} is inherently **serially correlated**, typically following structure $MA(h)$
  - While this doesn't affect the consistency of $\hat{\beta}_h$, it necessitates robust standard errors.

LP vs. VAR: A Comparison

- **VAR (Iterative):** Quality of h-step-ahead responses depends on the quality of the 1-step-ahead response, which is iterated forward.
- **LP (Direct):** Directly projects the outcome onto the space of variables at time t.
- **Key Findings (Plagborg-Møller & Wolf, 2021):**
  - Asymptotically, LPs and infinite-order VARs are **equivalent**.
  - LPs have a "doubly robust" property: they maintain better probability coverage even when the model is mildly misspecified.
- **Truncation Bias:** LPs have lower bias than VARs at long horizons when the lag length is misspecified.

Causal Inference: Baseline LP-IV

- **Motivation:** Macroeconomic interventions are often endogenous (e.g., central banks responding to inflation).
- **The Method:** Use external instruments (z_t) to isolate exogenous variation in the policy variable s_t.

Shocks in Practice: Literature Examples

- Monetary Policy:
  - Narrative: Romer and Romer (1989, 2004), extended by Wieland and Yang (2020).
  - High-Frequency: Gertler and Karadi (2015), Miranda-Agrippino and Ricco (2021), Bauer and Swanson (2023).
  - EU: Altavilla, Brugnolini, Gurkaynak, Motto and Ragusa (2019), Jaroncinski and Karadi (2020)
- Fiscal Policy:
  - Tax: Romer and Romer (2010), Mertens and Ravn (2013, 2014), Cloyne and Surico (2017), Cloyne, Martinez, Mumtaz and Surico (2022), Cloyne, Postel-Vinay and Dimsdale (2023)
  - Spending: Ramey (2011), Owyang, Ramey and Zubairy (2014), Ramey and Zubairy (2018)
- Others:
  - Supply: Oil shocks (Känzig, 2021).
  - Financial: Credit shocks (Gilchrist and Zakrejsek, 2012).
  - Macroprudential: Richter, Schularick, and Shim (2019).

Handling External Shocks: OLS vs. IV

Once an external shock ϵ_t is identified, it can be utilized in two primary ways:

- **Method 1: Direct OLS**
  - **Regression:** $\Delta_h y_{t+h} = \alpha_h + \beta_h \epsilon_t + \gamma_h X_t + v_{t+h}$.
  - **Key Assumption:** The shock is **observed without error**.
- **Method 2: LP-IV (2SLS)**
  - **Approach:** Use the external shock ϵ_t as an instrument (z_t) for an endogenous policy variable s_t. Must satisfy a strong first-stage ($F > 10$)
  - **Stock & Watson (2018) Conditions:**
    1. **Relevance:** $E(z_t, s_t) \neq 0$.
    2. **Basic Exogeneity:** Instrument is uncorrelated with the outcome's structural error.
    3. **+Lead-Lag Exogeneity:** due to dynamics $E(z_{t+l}, v_{t+h}) = 0$ for all $l \neq 0$.

The effects of an unexpected increase in FF Rate

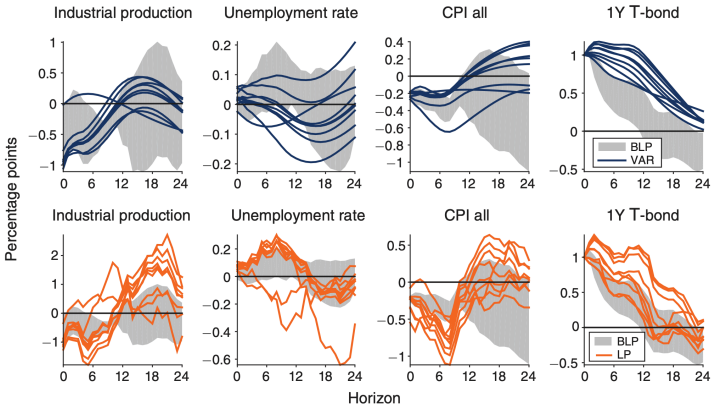


FIGURE 13. VAR, BLP, AND LP RESPONSES ACROSS SUBSAMPLES

Notes: Top row: VAR(12) (blue, solid) and BLP(12) (gray, area) responses. Bottom row: LP(12) (orange, solid) and BLP(12) (gray, area) responses. Subsamples 1982:2006, . . ., 1990:2014.

Figure 1: Source: Miranda-Agrippino and Ricco 2021

So You Want To Run a Local Projection? What Next?

1. Once I have my shocks, how should I use them?
2. Should I use levels or long differences?
3. What else should I control x for?
4. What standard errors should I use?
5. How can I assess the contribution of shocks to the variance of outcome variables?
6. How can I test and account for non-linearities?
7. How do things change if I decide to use panel data?
8. What non-OLS estimators can I use?

Techniques of Empirical Macroeconomics—Oscar Jorda and Karel Mertens

Appendix

The effects of an unexpected fall in ff rate: SVARs

- Huge question! (fed funds rate: overnight rate at the Fed)
- Sims (1980) developed a “simple” methodology that didn’t rely on very strong theoretical assumptions and that could offer an answer.
- Contents:
  - some very linear time-series econometrics
  - identification assumptions
  - illustration of results using “impulse response diagrams”
- Huge impact, other uses followed (effects of fiscal shocks, etc.).

A VAR

- A VAR(p) is a multivariate generalization of an AR(p) process. Example:

$$x_t \equiv \begin{bmatrix} \pi_t \\ y_t \\ i_t \end{bmatrix}$$

- A VAR is

$$x_t = \mathbf{B}_1 x_{t-1} + \mathbf{B}_2 x_{t-2} + \ldots + u_t$$

- $u_t \sim \text{iid}(0, \Sigma)$ is a vector of reduced-form innovations.
- Shocks are usually correlated and not directly interpretable. For example, a change in the interest rate may be a response to an inflation shock or to an output shock (or a combination)

Purposes of VARs

1. Data description — provide benchmark for macro models
   - Impulse responses.
   - Variance decomposition.
2. Forecasting — VARs provide good short-run forecasts, better than many other methods.
3. Structural interpretation and policy evaluation. Requires more assumptions — structure.

Identification Assumptions

- What is the impact of a increase in the fed funds rate?
- Cannot answer if everything contemporaneously depends on everything else without imposing some structure.
- Need to map u vector into a vector expressing "deep shock vector" ϵ
- Main structure: on contemporaneous relations, by using assumptions (based on "theory"). The first to be used was the triangular recursive identification structure.

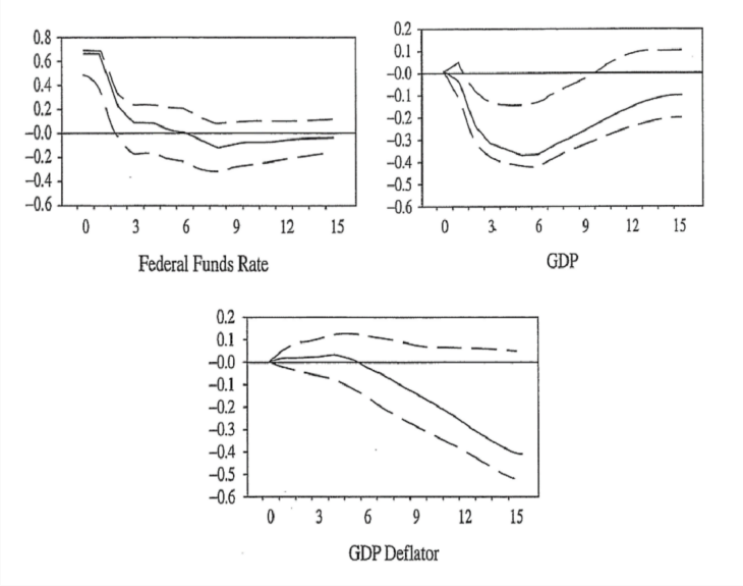
Triangular Example

- Model with inflation, ouput, and the interest rate. Assume: inflation does not respond to any of the other variables contemporaneously, output responds to inflation and the interest rate to both (e.g., Taylor rule).
- This amounts to:

$$\begin{bmatrix} u_{\pi,t} \\ u_{u,t} \\ u_{i,t} \end{bmatrix} = \begin{bmatrix} g_{11} & 0 & 0 \\ g_{21} & g_{22} & 0 \\ g_{31} & g_{32} & g_{33} \end{bmatrix} \begin{bmatrix} \varepsilon_{\pi,t} \\ \varepsilon_{u,t} \\ \varepsilon_{i,t} \end{bmatrix}$$

- Estimate reduced form of B, u, g can be computed and $[\pi_t, y_t, i_t]'$ can be expressed as function of fundamental shocks.

Identification Schemes

- The recursive assumption identifies the three shocks as an inflation shock, an output shock, and a monetary policy shock.
- Immediately we also obtain responses over time to these shocks, e.g., how the economy is affected over time by monetary policy.
- Other identification schemes:
  - some shocks have no impact in long run (Blanchard-Quah);
  - sign restrictions on responses (partial identification)

Impulse-response: effect of monetary policy shock